

	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^-}^{\#1} \alpha\beta_X$				
$\mathcal{K}_{0^+}^{\#1} \dagger$	$\frac{1}{2} (-3 Y_1 k^2 - 3 \overset{(2)}{\kappa}_3)$	0				
$\mathcal{K}_{0^-}^{\#1} \dagger^{\alpha\beta_X}$	0	0	$\mathcal{K}_{1^+}^{\#1} \alpha\beta$	$\mathcal{K}_{1^+}^{\#2} \alpha\beta$	$\mathcal{K}_{1^-}^{\#1} \alpha$	$\mathcal{K}_{1^-}^{\#2} \alpha$
	$\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha\beta}$	$\gamma_2 k^2$	$-\sqrt{2} \gamma_2 k^2$	0	0	
	$\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha\beta}$	$-\sqrt{2} \gamma_2 k^2$	$2 \gamma_2 k^2$	0	0	
	$\mathcal{K}_{1^-}^{\#1} \dagger^\alpha$	0	0	$\frac{1}{2} (\gamma_3 k^2 - 2 \overset{(2)}{\kappa}_3)$	$\frac{\gamma_4 k^2 + 2 \overset{(2)}{\kappa}_3}{2 \sqrt{2}}$	
	$\mathcal{K}_{1^-}^{\#2} \dagger^\alpha$	0	0	$\frac{\gamma_4 k^2 + 2 \overset{(2)}{\kappa}_3}{2 \sqrt{2}}$	$\frac{1}{2} (\gamma_5 k^2 - \overset{(2)}{\kappa}_3)$	$\mathcal{K}_{2^+}^{\#1} \alpha\beta$ $\mathcal{K}_{2^-}^{\#1} \alpha\beta_X$
				$\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta}$	0	0
				$\mathcal{K}_{2^-}^{\#1} \dagger^{\alpha\beta_X}$	0	$\frac{3 k^2 \overset{(4)}{\kappa}_{15}}{2}$

	$\mathcal{T}_{0^+}^{\#1}$	$\mathcal{T}_{0^+}^{\#1}{}_{\alpha\beta\chi}$				
$\mathcal{T}_{0^+}^{\#1}{}^\dagger$	$\frac{1}{\text{Det}(0^+)}$	0				
$\mathcal{T}_{0^+}^{\#1}{}^\dagger{}^{\alpha\beta\chi}$	0	0	$\mathcal{T}_{1^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{T}_{1^+}^{\#2}{}_{\alpha\beta}$	$\mathcal{T}_{1^+}^{\#1}{}_{\alpha}$	$\mathcal{T}_{1^+}^{\#2}{}_{\alpha}$
$\mathcal{T}_{1^+}^{\#1}{}^\dagger{}^{\alpha\beta}$	$\frac{1}{3 \text{Det}(1^+)}$	$-\frac{\sqrt{2}}{3 \text{Det}(1^+)}$	0	0		
$\mathcal{T}_{1^+}^{\#2}{}^\dagger{}^{\alpha\beta}$	$-\frac{\sqrt{2}}{3 \text{Det}(1^+)}$	$\frac{2}{3 \text{Det}(1^+)}$	0	0		
$\mathcal{T}_{1^+}^{\#1}{}^\dagger{}^{\alpha}$	0	0	$\frac{\gamma_5 k^2 - \kappa_3^{(2)}}{2 \text{Det}(1^+)}$	$\frac{-\gamma_4 k^2 - 2 \kappa_3^{(2)}}{2 \sqrt{2} \text{Det}(1^+)}$		
$\mathcal{T}_{1^+}^{\#2}{}^\dagger{}^{\alpha}$	0	0	$\frac{-\gamma_4 k^2 - 2 \kappa_3^{(2)}}{2 \sqrt{2} \text{Det}(1^+)}$	$\frac{\gamma_3 k^2 - 2 \kappa_3^{(2)}}{2 \text{Det}(1^+)}$	$\mathcal{T}_{2^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{T}_{2^+}^{\#1}{}_{\alpha\beta\chi}$
			$\mathcal{T}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta}$	0	0	
			$\mathcal{T}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta\chi}$	0	$\frac{2}{3 k^2 \kappa_{15}^{(4)}}$	

Abbreviations used in matrices

$$\begin{aligned} Y_1 &= \binom{(4)}{\kappa_1} + \binom{(4)}{\kappa_2} \&\& Y_2 = 2 \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_3} \&\& Y_3 = 3 \binom{(4)}{\kappa_{15}} - 2 \binom{(4)}{\kappa_2} \&\& Y_4 = 2 \binom{(4)}{\kappa_2} + \binom{(4)}{\kappa_7} \&\& \\ Y_5 &= 15 \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_2} - 9 \binom{(4)}{\kappa_3} - \binom{(4)}{\kappa_7} \&\& \text{Det}(0^+) = -\frac{3}{2} (k^2 (\binom{(4)}{\kappa_1} + \binom{(4)}{\kappa_2}) + \binom{(2)}{\kappa_3}) \&\& \text{Det}(1^+) = 3k^2 (2 \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_3}) \end{aligned}$$

Lagrangian

$$\begin{aligned} & \overset{(2)}{\mathcal{K}}_3 \mathcal{K}^\alpha{}_\beta \mathcal{K}^\chi{}_{\beta\chi} + \overset{(4)}{\mathcal{K}}_1 \partial_\beta \mathcal{K}^\delta{}_{\chi\delta} \partial^\chi \mathcal{K}^\alpha{}_\beta + \overset{(4)}{\mathcal{K}}_2 \partial_\chi \mathcal{K}^\delta{}_{\beta\delta} \partial^\chi \mathcal{K}^\alpha{}_\beta + \overset{(4)}{\mathcal{K}}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}{}^\delta - 6 \overset{(4)}{\mathcal{K}}_{15} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}{}^\delta + \\ & 4 \overset{(4)}{\mathcal{K}}_3 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}{}^\delta + 3 \overset{(4)}{\mathcal{K}}_{15} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^\delta - \overset{(4)}{\mathcal{K}}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^\delta + \overset{(4)}{\mathcal{K}}_7 \partial^\chi \mathcal{K}^\alpha{}_\beta \partial_\delta \mathcal{K}_{\chi\beta}{}^\delta + \\ & 3 \overset{(4)}{\mathcal{K}}_{15} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}{}^\delta - 2 \overset{(4)}{\mathcal{K}}_3 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}{}^\delta + \overset{(4)}{\mathcal{K}}_{15} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} + \overset{(4)}{\mathcal{K}}_{15} \partial_\delta \mathcal{K}_{\beta\alpha\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} \end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_0^{\#1\alpha\beta\chi} == 0$	1	$\partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} ==$ $\partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} + \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta}$	
$2 \mathcal{J}_1^{\#1\alpha\beta} + \mathcal{J}_1^{\#2\alpha\beta} == 0$	3	$\partial_\chi \mathcal{J}^{\alpha\beta\chi} + \partial_\chi \mathcal{J}^{\chi\alpha\beta} == \partial_\chi \mathcal{J}^{\beta\alpha\chi}$	
$\mathcal{J}_{2^+}^{\#1\alpha\beta} == 0$	5	$3 \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 3 \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \eta^{\alpha\beta} \partial_\epsilon \partial^\epsilon \partial_\delta \mathcal{J}^{\chi\chi\delta} == 2 \partial_\delta \partial^\beta \partial^\alpha \mathcal{J}^{\chi\chi\delta} + 3 (\partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi})$	
Total # constraint(s):	9		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$-\frac{21 \kappa_{15}^{(4)2} - 24 \kappa_{15}^{(4)} \kappa_3^{(4)} + 7 \kappa_3^{(4)2}}{22 \kappa_{15}^{(4)3} - 23 \kappa_{15}^{(4)2} \kappa_3^{(4)} + 6 \kappa_{15}^{(4)} \kappa_3^{(4)2}}$
	1	0	$\frac{88 \kappa_{15}^{(4)2} - 92 \kappa_{15}^{(4)} \kappa_3^{(4)} + 24 \kappa_3^{(4)2} - \sqrt{(11 \kappa_{15}^{(4)} - 6 \kappa_3^{(4)})^2 (105 \kappa_{15}^{(4)2} - 108 \kappa_{15}^{(4)} \kappa_3^{(4)} + 28 \kappa_3^{(4)2})}}{22 \kappa_{15}^{(4)3} - 23 \kappa_{15}^{(4)2} \kappa_3^{(4)} + 6 \kappa_{15}^{(4)} \kappa_3^{(4)2}}$
	1	0	$\frac{88 \kappa_{15}^{(4)2} - 92 \kappa_{15}^{(4)} \kappa_3^{(4)} + 24 \kappa_3^{(4)2} + \sqrt{(11 \kappa_{15}^{(4)} - 6 \kappa_3^{(4)})^2 (105 \kappa_{15}^{(4)2} - 108 \kappa_{15}^{(4)} \kappa_3^{(4)} + 28 \kappa_3^{(4)2})}}{22 \kappa_{15}^{(4)3} - 23 \kappa_{15}^{(4)2} \kappa_3^{(4)} + 6 \kappa_{15}^{(4)} \kappa_3^{(4)2}}$
	1	$-\frac{\kappa_3^{(2)}}{\kappa_1^{(4)} + \kappa_2^{(4)}}$	$-\frac{2}{3(\kappa_1^{(4)} + \kappa_2^{(4)})}$
	3	$\frac{6(11 \kappa_{15}^{(4)} - 6 \kappa_3^{(4)}) \kappa_3^{(2)}}{90 \kappa_{15}^{(4)2} + 36 \kappa_2^{(4)} \kappa_3^{(4)} - \kappa_7^{(4)} - 6 \kappa_{15}^{(4)} (11 \kappa_2^{(4)} + 9 \kappa_3^{(4)} + \kappa_7^{(4)})}$	$-\frac{2(306 \kappa_{15}^{(4)2} + 108 \kappa_3^{(4)2} + 12 \kappa_3^{(4)} \kappa_7^{(4)} + \kappa_7^{(4)2} - 8 \kappa_{15}^{(4)} (45 \kappa_3^{(4)} + 2 \kappa_7^{(4)}))}{(11 \kappa_{15}^{(4)} - 6 \kappa_3^{(4)}) (90 \kappa_{15}^{(4)2} + 36 \kappa_2^{(4)} \kappa_3^{(4)} - \kappa_7^{(4)} - 6 \kappa_{15}^{(4)} (11 \kappa_2^{(4)} + 9 \kappa_3^{(4)} + \kappa_7^{(4)}))}$
Resolved unitarity condition(s):		(Demonstrably impossible)	